1 Workplace and Resource Configuration

Overview

HYDRA menu	Master data → Resources → Resource configuration Master data → Workplaces/machines → Workplace configuration
FEDRA menu	Detailed Scheduling → Master data → Resource configuration
Transaction code	res
Function authorization	mdres mdresgenh for fields in combination with Test Equipment Management

Available user fields		
Where?	**Object type/user field key**	**Source (type)**
Tab *User fields*	<Res.type*)>/depending on data record	Resource (MF-D)
Table	RES/SYSTEM	Resource (MF-D)

*) <Res.typ> = resource type

The resource configuration is the central function to manage resources in the MES.

Purpose

This application manages the master data of workplaces/machines and other resources (tools, DNC resources, etc.). The resource type classifies resources. Each resource type is also linked to specific functions and applications, which provide further functionalities of the MES for resources of the specified type.

Integration

Use this application to view the resource information of all resource types available in the system. The resource type also specifies how and if data records can be edited. Depending on the resource type, you cannot edit all fields or create and delete all resources.

Based on the resource type, the MES also includes further applications that are especially tailored to these types. The *machine data collection* application package, for example, is based on resources of the type "machine".

In addition to the resource configuration, the *resource overview* application is available. You cannot use the *resource overview* application to edit data. This application only allows administrative operations for the daily handling of resources such as the stock transfer of resources.

Requirements

Create a year model/shift calendar prior to creating a workplace or machine. If you want to use the various resource types effectively, you also need the advanced licenses for these types.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can assign individual resource types to the other resources by configuration. Predefined resource types include:

DNC	NC/DNC program
DOC	Document
ENE	Energy meter
ENT	Removal device
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

We recommend using the predefined resource types.

The displayed detail resource information varies with the resource selected in the table overview.

Name

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

This detail application includes four main tabs:

- Resource configuration
- Resource list
- Resource attributes
- DNC versions

Main tab *Resource configuration*

Here, you can define the configurations and master data of resources.

General tab

Resource type

Resource type of the resource. The system delivery includes some default resource types. Create additional resource types in the application .

Resource

Enter the number of the resource or workplace to be collected in this field.

The resource type also specifies the maximum number of characters that are allowed for the resource number:

- Resources of the type MNR: a maximum of 8 digits
- Resources of a type <> MNR: a maximum of 20 digits

Permitted characters: ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890_.-+#. Do not use spaces and other special characters. For technical reasons, you can enter * (asterisk) and % (percent), but they are nonetheless not permitted because they are not valid characters. When you exit the input field, the system automatically converts lower case letters into CAPITAL LETTERS.

Please note for workplaces/machines (resource type MNR):
For technical reasons, the system does not check the maximum number of digits allowed for resources of the type MNR. For this reason, make sure that the resource number length (= workplace/machine number) does not exceed 8 digits.
Please note: If you set the resource type MNR before entering the resource ID (machine number), the GUI only allows you to enter eight digits.

If you select the option "numeric machine number" (basic parameter settings) for use with DOS terminals, you must ensure that the resource number (= workplace/machine number) only includes numerical digits and that its length is exactly 8 digits. If necessary, prefix leading zeroes to the number to extend it to eight digits, when creating the workplace/machine.

Short name

Short name of the resource. Only use this field with workplaces/machines (resources of the type MNR).

Name

Use this field to assign a short, unique name to each resource. Reports and overviews as well as terminal dialogs show this name, which is also useful for orientation purposes.

Responsibility area

Use responsibility areas to restrict the data users can view in different evaluations/reports. Users can only view the data they are allowed to according to their responsibility area authorization.

The *responsibility area* field can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.

If you leave the *responsibility area* field empty, the system automatically enters the value "--DEFAULT--" in the field. Resources including this value are always displayed regardless of the user's assigned responsibility authorizations.

Cost center

This field includes the cost center the resource is assigned to.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information in form of comments.

Acquisition date, acquisition costs

Additional information in form of comments.

Configure the currency for the entire system in the basic settings.

Storage location

Location where the resource is stored when it is not being used (original storage location).

In connection with the Material and Production Logistics (MPL) product group, this field specifies a material buffer. If you log on an input batch, the logged on input batch(es) will be transferred from the previous material buffer to the material buffer entered in this field (upstream of the machine).

Delivery date, start-up date, guarantee date

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

External designation, resource type designation, usage, purchase order number

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Supplier and party in charge including detail fields

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Workplace configuration tab

This tab is only available if you select a resource of the type "MNR".

Workplace master data

Workplace category

N Machine

P Workplace

Defined as machine or workplace. If you exclusively use BDE and/or MDE and PDV, the two categories are identical as regards processing.

J Machining center (BDE-BEA only)

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined as mere CAQ inspection station and does not affect BDE or MDE statistics.

R Coil-based manufacturing (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

S Cutting unit (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

D Parallel output batches (only MPL)

You can produce parallel output batches on the machine for an operation that requires batch management.

C Packing station (only MPL)

You can use specific posting functions of the machine to represent a packing station. The functions are described in detail in the AIP-LCS product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

F Laboratory/in-production inspection

This workplace is configured as inspection station. The inspection points are displayed, which are assigned to this workplace or machine group of this workplace because of the higher-level inspection point. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

W Goods receipt inspection

This workplace is configured as inspection station. The goods receipt inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

K Calibration

This workplace is configured as inspection station. The calibration inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

Workplace type

E Single workplace (SWP)

G Group workplace (GWP)

Group workplaces are workplaces without machine data collection or MDE evaluations.

In case of group workplaces, you cannot post to resource performance accounts in an operation-related manner with postings based on the current machine status. Only main production times (RPA 11) are recorded. You must define a status with the control indicator "production" in the .

The system does not generate for group workplaces. Therefore, MDE evaluations that evaluate MDE log records are not possible.

Like single workplaces, you can assign group workplaces to terminals. In this case, you have to make sure that the is set to operation mode "BDE" or the option *Processing* is set to "BDE processing" in the .

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this option is checked, the machine/workplace has been (logically) deleted. In this case, the system does no longer permit the following changes:
- Order postings on the terminal
- Order postings on the MOC (e.g. using the "order overview" function)
- Changes when editing events

The graphic planning board of the Shop Floor Scheduling and the application *Workplace assignment* do no longer show the machine/workplace.

Blocked machines/workplaces are shown in evaluations and overviews. If blocked machines/workplaces are not shown, this is then described in the relevant documentation of the MOC application.

Tip: In applications where data is selected according to the responsibility area authorization, you can hide machines/workplaces if you remove the responsibility area.

Company

Use this field to differentiate the individual machines/ workplaces. The system can use this field for evaluation purposes.

Group

Use this field to assign the workplace/machine to a logical group. In planning, this is a capacity group. Capacity groups combine primary capacities.

If you create a new workplace, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If the capacity group does not yet exist, the system automatically creates a capacity group and assigns the workplace.

Category

Enter the category of the machine. By means of this, you can enable a validation check according to the BDE configuration: *Master data > Order configuration > Order types,* tab *validation,* option *Check planned workplace/group/category on OP logon* (value *category).*

Year model

Enter a valid year model . The times to be posted are compared with this shift model when they are recorded. If you have not defined a planned year model in the HLS tab, the shift model entered here is also used in the Shop Floor Scheduling.

Standard rate, machine

Enter the arithmetical standard rate of machines for calculations. The Shop Floor Scheduling uses this value for some (evaluated) KPIs.

Standard labor rate

Enter the arithmetical standard labor rate for calculations. The Shop Floor Scheduling uses this value for the KPI "Evaluated labor utilization".

Performance level

You can enter the performance level of the workplace/machine in percent in this field. The Shop Floor Scheduling and the evaluation of material requirements integrate this value when calculating the remaining run time.

Incentive wage indicator

Defines the type of calculation used for incentive wages. This option is mostly used in combination with the incentive wages based on formulas for customer-specific configurations. In addition, use the "incentive wage indicator" as selection criterion for the *wage type determination* to calculate incentive wages.

Leave this field empty, if you do not use the incentive wage module.

The *incentive wages indicator G=group calculation* has a special meaning. If this option is set for a workplace/machine, you have to assign a premium group every time you log on an order. You can do this either via
- the "assignment of premium groups" option of the product group *Incentive wages* or, optionally, via
- an additional field in the terminal dialog for the logon of orders. If no assignment is available, the system rejects the logon of the order by issuing a validation error.

Therefore, you may only assign the incentive wage indicator *G = Group calculation,* if the group premium conditions are met in the incentive wages calculation, as otherwise orders can no longer be logged on!

You can specify the meaning of the other incentive wage indicators according to your requirements while customizing the system.

File

You can assign a graphic to each machine/workplace. The *workplace overview* or the AIP shows this graphic, for example. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.

In the path configuration, you must have configured the following:
- the path ID "MOCWPIMG" for the MOC or SMA
- the path ID “HYDRA” (also see) for the AIP. The file name length of graphic files is restricted to 12 characters (8.3 notation). Note for Linux installations: only use lower case letters for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, define the maximum capacity in KG here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Data collection

Display 3rd list

Use the options described here to show/enable a third list in the main view of a Windows terminal (CTWIN / AIP). You can switch between the respective terminal lists depending on the options set. The following settings are possible. Please note that the contents displayed in the lists depend on the product group in use:

→ Input material (MPL): shows logged on input materials/ batches.

→ Resources (WRM): shows logged on resources and tools.

→ Staff (BDE): shows logged on staff.

→Output material (MPL): Produced output batches are displayed.

Show material/PRT list when OP is logged on

This option is only relevant in connection with the WRM module and the resources logged on to the Windows terminals (CTWIN / AIP).

If this option is set and you log on an OP, a specific login dialog opens. This dialog includes a list of components/production resources and tools. This list shows resources that meet at least one of the following requirements:
- the option "posting to terminal" is set in the resource type;
- the option "log on with OP" is set to "explicit logon" for the resource.
- the resource is a so-called "required resource" (option is set for the resource).

Please note: If the workplace is relevant for MPL, the list also shows material components.

Sequencing list

This option defines which operations are displayed in the sequencing list of the terminal. The following settings are available:

S Basic setting. The system takes the value from the option of the same name in the *HYDRA basic settings*.

M Pool of workplaces. The terminal sequencing list only shows the operations planned for the workplace.

G Pool of workplaces and groups. The terminal sequencing list shows operations that are:

- planned for the current workplace or
- for another workplace of the group or
- that are still located in the pool of groups.

K Pool of workplaces and categories. The terminal sequencing list only shows the operations that are planned for workplaces of the selected category.

H Group control. The terminal sequencing list shows the operations that are
- planned for the current workplace or
- for another workplace of the group.

Number of OPs in sequencing list

Enter the maximum number of operations that are to be displayed in the terminal sequencing list. Enter 0 if you want to show all operations.

Compulsory sequence

Use this option to specify if it is mandatory to log on the OPs in the planned sequence. The following parameters are permitted:

N Disabled

J Enabled

If the parameter is "enabled" and you log on an OP, the system checks whether the order backlog for this machine/workplace includes an OP that is planned for the same time or previous to this OP, but has not yet been started (i.e. status = V/prepared). If yes, the system rejects the logon of this OP.

Note: If you plan orders in the system using the *Order sequencing* (menu Production control → Production support → Order sequencing) and you configure the sequencing list with any other option than "M" (pool of workplaces) and you enable the *compulsory sequence,* this might lead to a combination that does not make sense.

Please note for the sequencing list:

- If the sequencing list includes operations that are in the status "interrupted", you can log on these OPs at any time, irrespective of the specified compulsory sequence.

Dialog control

To meet this requirement, define a dialog control that deviates from the standard behavior for the workplace in the dynamic dialog configuration of the Windows terminal (CTWIN / AIP). Then refer to the dialog control in the dialog.

Use this configuration only as part of customizing the HYDRA system. Otherwise the configuration is not relevant.

Logon of several OPs

Select this option, if several different operations should be processed on the machine. Otherwise, the system only allows one operation to be logged on to the machine.

Possible values:

Y	Log on as many OPs as required at the same time. Please note: The system allows a maximum of 20 operations to be logged on simultaneously to a machine, if the machine is assigned to a terminal with operation mode *MDE.* If more than 20 operations must be logged on at the same time, MPDV must review the conditions in order to remove the limitation. If MPDV agrees to remove the limitation, you can do so, otherwise search for alternative solutions. MPDV analyzes the conditions as part of a service.
N	You can log on one OP only.
1...9	You can log on a maximum of n OPs.

Posting

Quantity posting to staff

Use this function to post the quantity of order interruptions/logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to staff can be found .

Posting for OPs that are not logged on

Use this option if you want to
- interrupt
- finish
- report part quantities for

operations that are not logged on to this workplace.

If you record quantities for an operation that is not logged on, the system posts these quantities onto the operation in the BDE module. The MDE module does not post the quantities.

If you want to use this function with the AIP terminal, the BDE posting dialogs that are installed by default require the following:
- use the simplified BDE posting dialogs (the so-called "") or
- customize the dialogs.

Then you will be able to enter an operation that is not logged on.

Posting of machine time with simultaneously logged on operations

If this option is set and OPs are logged on simultaneously, the system posts the machine time proportionately onto the operations.

Y	Proportionate posting on OP according to the number of OPs
N	No proportionate posting. If the option is not set, the complete machine time is posted for each operation.
V	According to the default quantity of the OPs. Make sure that the default quantity (target quantity in primary quantity unit) of the operation is > 0.
Z	According to the standard time of the OPs. Make sure that the standard time (processing time) of the operation is > 0.

Please note:

This option is also evaluated for group workplaces and in general you should better not use this option for group workplaces.

Automatic logoff of staff when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Use this option to configure the personnel-related data collection at MDE workplaces. If you use HYDRA MDE, the terminals can generate fully automatic shift ends. You can configure here if
- the staff logged on to the workplace should be logged off automatically at the end of the shift or
- if they should remain logged on.

Y	Always log off staff when the shift ends.
N	Always save staff when the shift ends except for manual logoff.

X Evaluate the person's settings. The system searches for the corresponding settings of the person .

Automatic OP posting when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Y Interrupt and log on again at beginning of shift

N Interrupt

Shop Floor Scheduling

Find further information about the HLS product group in the relevant HLS documentation.

Planning function

This option specifies whether a workplace or a machine will be displayed and if so, in which planning function.

P Planning in the graphic planning board of the Shop Floor Scheduling or in the graphic order sequencing (GAV), i.e. you plan the workplace via the Shop Floor Scheduling or the graphic order sequencing; the workplace is then displayed in these applications, but not in the tabular order sequencing (AVG).

Note: There are also other settings that specify whether a workplace is displayed in the Shop Floor Scheduling or in the graphic order sequencing:
- the workplace must be assigned to a group identified as a "capacity group"
- you must be authorized for the responsibility area of this workplace
- planning profile

H Only relevant, if you use the HYDRA Shop Floor Scheduling module (HLS).

Like P.

T Reserved

A Planning in the tabular order sequencing (AVG), i.e. you plan the workplace using the AVG product group.

N No planning; the tabular order sequencing (AVG), the graphic order sequencing and the HLS module do not show the workplace.

Planned year model

Here, you can enter a special year model only used for planning in the Shop Floor Scheduling. This year model does not affect data collection and posting in the product groups BDE/MDE. If you do not define a planned year model, the system uses the year model (*Master data* tab) for the planning.

Availability

Define the available capacity of a workplace/machine. The default value for the available capacity is 1000 [per mill].

In the Shop Floor Scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. exactly one operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs you about the double assignment. If you use the automatic assignment, multiple assignments are generally not feasible.

Use this setting to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, set the available capacity to 2000 [per mill] in this field.

If nothing is entered in this field or if you enter the value 0, the system interprets this as the default value of 1000 [per mill].

This functions requires a corresponding license.

Check personnel availability

Choose from the following options:

- Check if at least one person is planned
- Check personnel availability
- Check personnel availability and qualification

When you operations in the , the system checks whether persons are planned in the application for the time of the scheduling You will find further information on the display of personnel capacities in the Graphic Planning .

This option is only available if you enable the extension .

MPL

For further information on the MPL product group, refer to the relevant MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the terminal posting dialogs. Possible values are:

N No batch processing

L Batch tracing (input/ output batches) as part of HYDRA MPL/TRT

D Throughput batch processing as part of HYDRA MPL/TRT

J Individual batch tracing (CHV)

The following functions are only available in connection with the product group Material and production logistics and are supported only by Windows terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

Subsequent material buffer

If you specify a material buffer in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log off operation) is automatically populated with this value.

If you do not enter a material buffer in the input dialog (e.g. deleted from the input field), the system automatically posts the output batch to the material buffer specified in the "subsequent material buffer" field.

Automatic generation of batch number

If you set this option, the system automatically generates a batch number for the output batch to be produced. Otherwise, the system expects you to enter the batch number for the new output batch to be produced, when you log on an operation or change the output batch.

Please note: If, in the field *Batch management* you set the option D (= *Throughput batch recording*), the system automatically sets the value for the *Automatic generation of batch number* to "J". In this case, you cannot enter the batch number manually.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. This option is only activated, once you have enabled the consumption balance for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport starts from the material buffer where the output batch is included. The configurations of the material type override the corresponding options of the resource configuration.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when you plan an operation for a machine via the Shop Floor Scheduling module. Transport starts from the output material buffer of the preceding operation. The configurations of the material type override the corresponding options of the resource configuration.

Quantities tab

This tab is only available if you select a resource of the type "MNR".

Conversion factors for base quantity

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts. In general, the system supports the following **quantity accounts:**

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported with each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The system design specifies the use of several quantity types or accounts. For example: If you want to enter the rework quantity manually, a corresponding input field must be configured in the input dialog (customization).

Use the quantity type "primary quantity" if you want to collect quantities automatically.

Quantity units and conversion factors for base quantity

Define a quantity unit for each quantity type. Use the alternative quantity accounts to enter data/quantities manually. In this case, the system does not convert quantities automatically.

If you do not enter data manually in the alternative quantity accounts, the server converts the quantities into the alternative accounts using:
- the conversion factors or
- the units that are configured in the MOC machine master data.

For further information on the conversion of quantities and examples, refer to the document .

Basis for HYDRA-MDE quantity conversion

Define the basis for the quantity conversion.

A Use the conversion factors of the OP that is logged on. If no operation is logged on, the system uses the quantity conversion stated in the machine/workplace configuration.

M Use conversion factors from the workplace configuration for the quantity conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit you want to use for data collection at this machine/ workplace. If you collect quantities automatically, these quantities are generally primary quantities.

If you want to convert quantities automatically into another quantity type, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Indicate the secondary quantity unit you want to use for posting the quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Indicate the tertiary quantity unit you want to use for posting quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Indicate the base quantity unit you want to use for posting quantities to the workplace/machine.

Manual entry of quantities, yield

Manual entry of yield

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting yield as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, scrap

Manual entry of scrap

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting scrap as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, rework

Manual entry of rework quantity

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note
Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting the rework quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, open quantity

Manual entry of open quantity

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting open quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

"MDE configuration" tab

This tab is only available if you select a resource of the type "MNR".

Monitoring

Monitoring type

Choose from the following monitoring types:

Monitoring via operating signal

No monitoring

Cyclic monitoring

If you select **cyclic** or **operating signal** monitoring, you can only enter a malfunction if the terminal prompts you to do so ("Assign malfunction"). If you do **not** use **automatic monitoring**, you can enter a new machine status at any time.

If you use the **cyclic** monitoring option, the machine automatically switches to the "production" status when counting pulses occur. If you select the "operating signal" option, the machine automatically switches to the status "production" as soon as the operating signal is set. If you do not use the "automatic monitoring" option, you must assign the "Production" status manually.

Entry of disturbance reason required with specified delay time in [s]

You can only use this function, if the following requirements are met:
- it is a Windows terminal (CTWIN, AIP)
- The Process Communication Controller (PCC) does not run in stand-alone mode.

If the system identifies a downtime without a reason, the terminal opens the input dialog "Change machine status" after the specified delay time. If the terminal goes back into production, the window still remains open.

If you now enter a machine status (during production), this data input activates a transfer posting event that changes the most recently recorded status from "General disturbance" to the newly entered status. If this change is ok, the window closes; otherwise, it remains open.

However, if the system identifies the next downtime (with or without a reason), you can no longer change to the previously noted status. The window closes automatically.

If the system identifies another downtime without a reason and the delay time has expired, then the input window opens as described above.

If the system identifies a downtime without a reason and the machine switches to production before the delay time expires, then the terminal does not automatically prompt you to enter a malfunction reason.

Important note:
This change only affects the HYDRA Machine Data Collection. The system does not correct the resource performance accounts of the currently running OP online!

Please note for data maintenance:
The tabular event maintenance of the MOC shows all changed machine statuses. However, you cannot edit the transfer posting event as it is locked. In order to perform recalculations correctly with respect to orders and machines, change the original event with the status "NOT ASSIGNED" to the correct status. The transfer posting event does not affect recalculation!

Minimum malfunction time

Specify a time in seconds for the minimum malfunction time. This value defines the time that a malfunction/disturbance must continue before the machine changes from the status "Production" to the status "Not assigned".

If operating signals are monitored, the status is directly changed. You can use the following explicit option in the MDEB2.ini to disable this behavior (deactivation of direct status change). Result: the status is only changed when the **minimum disturbance time** has expired:

MDEB2.INI

```
[INIT]
;Activating the direct status change (globally or for a specific machine)
SetMStatusDirect=1
SetMStatusDirect@<machine number>=1

;Deactivating the direct status change (globally or for a specific machine)
SetMStatusDirect=0
SetMStatusDirect@<machine number>=0
```

Minimum cycle time

If you select the **cyclic** monitoring option, specify a minimum cycle time in seconds in this field.

The terminal uses this minimum cycle time and the target cycle that is stored with the (logged in) operation and that is set off against the cycle extension to calculate the maximum value. The terminal uses this maximum value as the default cycle time.

If both, the minimum cycle time and the target cycle stored for the operation, are 0, the default cycle time is set to 60000 seconds [per 1000 machine clocks].

Cycle extension

If you select the **cyclic** monitoring option, enter the percentage for extending the target cycle time in this field. Enter a value ranging between 0 and 5000.

The system offsets the target cycle stored with the (logged in) operation against this percentage. A value less than 100 is a shortened cycle; a value greater than 100 is an extended cycle.

Number of target cycles

If you select the "**cyclic** monitoring" option, enter the number of cycles (0 to a maximum of 9) after which the terminal automatically switches from a status unequal to "production" into the "production" status within the cycle time (requirement: the status that is unequal to production is not locked for the "production" status).

Some production processes provide machine cycles during the setup phase. Set a value greater than 0 in order to prevent the current machine status from changing immediately. Please note: The quantities you collect until the machine switches to the "production" status are neither posted as yield nor scrap.

Cycles to be evaluated

Reserved Enter 0 in this field.

Management

Posting during production lock

Use this setting to specify how to post the counting pulses that are collected while the status "production" is suspended. This configuration takes effect for all counters configured as "Yield".

Posting as scrap If this option is configured for the counter, the system offsets the counting pulses against the partitioning/ pulse factor and posts these pulses as scrap. Even if you defined another quantity account for offsetting, this one will not be used.

Posting as yield parts the system posts the counting pulses as yield

No posting the system does not post the quantities while the "production" status is suspended.

Pulse factor specific to machines

Use the pulse factor, for example, if you want to collect lengths (e.g. using a wheel).

Set the value to 0 for machines where a discrete or integral number of quantities (e.g. pieces) is collected per pulse. In this case, the pulse factor is not evaluated. That means, the number of cycles posted corresponds to the actual pulses transferred via the MSS (machine interface).

The MSS (machine interface) records the signals transferred from the machine (counting pulses). According to the configured number of pulses, the system calculates and posts the quantities as follows:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

The system interprets pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock* > scrap) as scrap. Also use the above-mentioned formula to calculate the scrap quantities.

Partitioning specific to machines

Enter the partitioning specific to the machine in this field. Multiply the machine-specific partitioning by the partitioning stored with the operation in order to integrate the machine-specific partitioning into quantity calculation. Enter the value 1 in this field, if you do not want this to happen.

Extended weekend automatic

If you select this option and the system is configured accordingly, the system assigns at the beginning of the shift the status that was available before status 999 was activated.

Note:

To use this option, the workplace must already be assigned to a terminal.

Find detailed information about the automatic activation of status 999 in the document .

Waiting period, short-term disturbance

Configure a *short-term disturbance* status for each machine/ workplace to improve the overview, e.g. in the machine history. Use this status as a “repository” for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal automatically identifies a downtime and the machine automatically goes back into production, the system checks if this disturbance is shorter than the time period configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

Enter the logical output where a digital signal should occur when the corresponding status is available.

Machine lock output	The system sets this output, if you enabled the option "set machine lock output" in the current machine status.
Target quantity reached output	The system sets this output, if the collected yield reaches the target quantity of the OP.
Machine downtime output	The system sets this output, if the machine is in a status unequal to *Production.* When changing to the *production* status, the system sets the output back to 0.
Free I/O	Free input/ output for customizations.

Use these statuses for connecting a monitoring light or a horn, for example.

Enter the corresponding number in one of the fields in order to assign an output and to specify which relay is interconnected by the terminal when the predefined status occurs. Enter "0" to prevent any action. Note that you cannot assign a terminal output more than once.

Please note

Specify the statuses that trigger the activation of the machine lock in the *Status assignment*.

Generally, enter the value "1" in the input field, when the machine lock is activated via the available relay output of a DS 100. In this case, the system sets the machine lock if
- a correspondingly configured status occurs <u>and</u>
- the status is *<u>not assigned</u>*.

Output batch change**

Customer-specific assignment of an input with an automatic output batch change (MPL). By default, enter 0 in this field.

PDE (Process Data Collection)

Collect process data

This parameter specifies if the system collects process data for this machine. If this parameter is not set for a machine, you cannot collect process data for this machine.

External connection

The AIP 8.2 and/or the PCC in stand-alone mode (MDE-Blade 2 Version 8.1.0.1) do no longer support the options marked with **. As they use other configurations for the connection.

External connection

If this machine is assigned to a master terminal the following connection options are available:

No external device	External devices are not connected
DS100	DS100 connection
Arburg control system**	Arburg connection
Engel interfacing**	Connection of Engel machines
MT3**	MT3 connection
PDE**	Process data collection

If you activate a DS100 or MT3** connection, you can select the field "device address". If you activate the option "Engel interfacing", you can select the field "serial number". If you activate the option "Arburg server system", you can select the field "class".

Note regarding the combination of connections on a master terminal:
"DS 100" and "No external device": allowed
"MT 3" and "No external device": allowed
"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)**

Enter the serial number of the connected Engel machine. Set the option "EMS machine interface" in the HYDRA basic parameter settings if you want to use Engel machines.

Device address

You can select this field, if you activate a DS100 or MT3** connection. Enter the device address of the sub-bus participant.

"Resource configuration" tab

For resources of type "MNR", only the fields marked with "*" are available:

- Family (section *resource master data*)
- Cycles (section *target utilization*)
- Runtime (section *target utilization*)

Resource master data

Type

Identifies the type of resource:

Resource: A resource can be uniquely identified, i.e. the resource is actually present. Its quantity is always 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then you can change the value in the field *Number* from 1 to another positive integer value. You cannot post data onto anonymous resources because anonymous resources do not relate to one specific resource.

Required resource: A required resource stands for one or more actual resources that can be identified. Specify in the configuration *WRM: Master data* > *Required resources* which resources are represented by a required resource. The number results from the number of actual resources assigned to the required resource.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future modifications.

Version

Revision number; store here the program version for resources of the type DNC.

Quantity

You can only edit this field, if it contains an anonymous resource and the option *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is calculated automatically for required resources.

Family*

Assign a resource family. If you change the resource family subsequently, an information dialog appears as a warning because user fields might possibly be assigned via the resource family.

Target utilization

Cycles*

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime*

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Input unit

Input unit

Absolute value limit (EMG 8.1, function authorization: resablim)

Enter the absolute value limit of the (meter) resource. The energy monitor shows this limit value in addition to the current meter reading. Use the Escalation Management to generate an escalation message, if the counter value of the resource exceeds the specified absolute value limit. You need the function authorization "resablim" to view this field.

Actual utilization

The periods when a resource was logged on to a workplace are the basis for posting the *cycles (clocks), runtime, yield,* and *scrap* as *actual utilization.*

Clocks

The cycles (clocks) posted for the resource up to now.

Runtime

The total time in hours posted for the resource up to now. The total time is the sum total of all times posted onto RPA 1 to 11.

Yield (B)

The yield posted for the resource up to now (base quantity unit).

Yield (P)

The yield posted for the resource up to now (primary quantity unit).

Scrap (B)

The scrap posted for the resource up to now (base quantity unit).

Scrap (P)

The scrap posted for the resource up to now (primary quantity unit).

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Original partitioning

Partitioning of the tool (= number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This value can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with one cycle/clock due to a tool defect.

Always use the current partitioning to post cycles to the tool.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection via the terminal.

Partitioning due to cavities

If you set the option "partitioning due to cavities", the system (re-)calculates the fields "current partitioning" and "original partitioning" using the values defined in the cavity management. Then, you can no longer change the fields manually.

Log on with OP

Use this option to specify whether or not you want to log on the resource with the OP. To do so, the resource must be included as a component in the operation's list of production resources and tools. Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource manually (explicitly) nor change the logon.

Explicit: You can manually (explicitly) log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource or another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Please note:

If you log on another resource explicitly (manually), this resource will be logged on for the resource that has the same *resource type* in the operation's list of production resources and tools. For this reason, you can only log on those resources explicitly (manually) that are included as a requirement in the operation's list of production resources and tools. In this way, you cannot log on a resource that is not included as a requirement in the list of production resources and tools (the resource must be entered in the list).

In general, you should not enable this option for the resource type DNC. The DNC product group handles this differently (NC programs are logged on separately).

The system also logs on resources that are defined in the BOM of the machine.

Parallel logon/ planning possible

You can log on/plan the tool simultaneously.

Please note: You can only log on a resource to one machine more than once. Consequently, the option "Parallel logon possible" refers to several different OPs logged on to one machine.

In this case, the system posts data proportionately as follows:

- Post quantities proportionally.
- Post times 100% for each resource. This means that the system posts double the time to the resource, if the resource is logged on twice.

Post to resource

Specifies whether or not the quantities and times are posted to the resource. Due to a high degree of complexity, you should only assign this option to those resources that you actually want to evaluate.

Collective lock

If you lock a lower-level (assigned) resource using the BOM function, the system sets a collective status for the higher-level resource. If this collective status is set, the system treats the higher-level resource as locked when a download request is made.

If you enable this function, the system passes the collective lock to the higher-level resource.

Planning

Setup time

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Teardown/retooling time

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

Not used. The system uses the configuration option of the same name stored in the resource type to integrate the resource allocation in the HYDRA Shop Floor Scheduling.

Evaluation

Integrate in evaluations

Reserved for future modifications.

File

File exists

Shows whether or not the file is stored in the specified path. A cyclic process checks the files and sets the options subject to whether or not the file is available.

File name

File name; without file extension for DNC. The system adds the file extension according to the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Enter two comparison resources for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resource to be compared.

Resource type 1

Resource type of the resource to be compared.

Resource 2

Resource number of the resource to be compared.

Resource type 2

Resource type of the resource to be compared.

Accuracy

Enter more detailed information on measuring accuracy and measuring range for test equipment resources.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

The workplace and resource configuration provides data of two basic object types. You can also edit this data in the workplace and resource configuration: on the one hand these are machines and workplaces and on the other these are the resources. Machines and workplaces are also "resources". But resources are not automatically machines and workplaces.

Object type

The system configures the user fields of machines/workplaces in relation to the object type "MNR". The system stores data contents to the machines/workplaces table and the resources table of the database to ensure data consistency.

The system configures user fields for resources in relation to the object type matching the resource type of the resource (example: create resources of the type "PAC" in relation to the object type "PAC"). The system stores data contents to the the resources table of the database.

User field key

Each user field key describes a combination of user fields. The management of the user field key (and therefore the meaning of the fields) is different for each object.

User fields

The following user fields are available after configuration:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16

Field data type	Number of fields
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. Configure the system accordingly to support this kind of user fields.

As the table shows resources of different types, use the user field key "SYSTEM" of the object "RES" to identify the column headings for the user fields.

Comment tab

Store additional resource comments in the "comment" tab.

Main tab *Resource attributes*

Shows additional resource attributes via the user field definitions of the resource family. Use the "resource attributes" button for editing.

Main tab *Resource list*

Shows the resource list for the selected resource. Click the "resource list" button to go directly to the BOM application for editing purposes.

Main tab *DNC versions* (available as of DNC 8.2)

Shows the available versions of a DNC resource including a flag indicating the currently applicable version. HYDRA provides this valid version for machine downloads.

Toolbar

General tab

Insert

Function authorization: mdres.create

Opens the dialog for adding a resource. This dialog provides the fields that match the selected resource type.

Copy

Function authorization: mdres.copy

Opens the dialog for copying an existing resource. Subject to the selected resource and its resource type, the copy function differentiates the following:

- Copy function for resources of **resource type = MNR** (workplaces, machines)
- Copy function for resources that do **not** have the **type MNR**

Copy function for resources of resource type = MNR (workplaces, machines)

From: resource type, resource, short name, name

- Resource type (fixed "MNR“)
- Workplace/machine number
- Short name
- Name

of the workplace you want to copy. You cannot change these values. They derive from the selected data record.

To: resource type, resource, short name, name

- Resource type (corresponds to the resource type of the workplace you want to copy; cannot be changed).
- Workplace/machine number
- Short name
- Name

of the target workplace.

Copy machine status

Function authorization: mdmst.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy counter configuration

Function authorization: mdctr.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Note that the counter numbers of the new workplace are identical with the counter numbers of the workplace you copied. If necessary, you have to adjust the counter numbers.

Copy reasons

Function authorization: mdreas.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy function for resources that do not have the resource type MNR

The copy function for all resources that do not have the type MNR opens the "insert" dialog and takes over the details from the previously selected resource. But you can edit and change all fields.

Edit

Function authorization: mdres.edit

Opens the dialog to edit a resource and provides the tabs and fields of the relevant resource type.

As of MES Weaver 4.0$_{pe}$, you can change master data of several selected resources of the same resource type at the same time. You can select up to 10 fields and assign a value. You require the function authorization ***mdresmm*** to edit several resources at once.

Delete

Function authorization: mdres.delete

Deletes one or several selected resources.

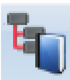
Resource tab

Configuration – resource status

Opens the application "resource status" to define statuses for all resources that do not have the type MNR.

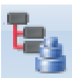
File - show file

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type.* And only available if the relevant license and function authorization are available.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – required resources

Opens the "required resources" application. The selected resource is entered as default value for the required resource.


Go to – cavity assignment

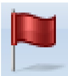
Opens the "cavity assignment" application. The selected resource is entered as default value.


Go to - resource attributes

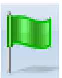
Opens the application "resource attributes". The selected resource is entered as default value.

Functions – Measures

Opens the *Measures* application.


Functions – Status change

Opens the dialog to change a resource status. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.


Functions – Release of resource

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.


Functions – Stock transfer

Opens the dialog to transfer/relocate a resource.

***Workplace* tab**


Configuration – status assignment

Opens the application "status assignment". The system enters the selected resource in the corresponding field.


Configuration – counter configuration

Opens the application "counter configuration". The system enters the selected resource in the corresponding field.


Configuration – terminal assignment

Opens the application "terminal assignment". The system enters the selected resource in the corresponding field.

 **Entry – reasons**

Opens the application "reasons". The system enters the selected resource in the corresponding field.

 **Entry – Operator positions**

Opens the application "operator positions". The system enters the selected resource in the corresponding field.

 **Entry – premium indicator**

Opens the application "premium indicator". The system enters the selected resource in the corresponding field.

 **Groups - groups**

Opens the application "groups". The system enters the group of the selected resource.

 **Groups – group assignment**

Opens the application "group assignment". The system enters the selected resource in the corresponding field.

 **Miscellaneous – cycle parameter**

Opens the application "cycle parameter". The system enters the selected resource in the corresponding field.

 **Miscellaneous - workforce requirements of workplaces**

Opens the application "workforce requirements of workplaces". The system enters the selected resource in the corresponding field.

***DNC* tab**

The tab is only available, if you select a DNC resource. These are resources configured as resources with DNC processing in the resource type.

 **Configuration – resource status**

Opens the "resource status" application.

Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".

Copy resource attributes (as of DNC 8.2)

Copies values of resources attributes from one resource to another. Both resources must use the same user field key.

File - comparison editor

Opens the comparison editor for the selected resource or resources. See below for further information.

File - export

Exports the file specified for the resource. You use the file explorer to specify the target file.

File - import

Imports the file specified for the resource. You use the file explorer to specify the source file.

File - viewer

Opens the file specified for the resource using the defined viewer program.

File - editor

Opens the file specified for the resource for editing using the defined editing program.

Set valid version (as of DNC 8.2)

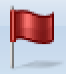
Only active, if you select a version in the *DNC versions* tab. The selected version is set as the new and valid version.

Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.

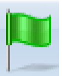
Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Functions – Status change

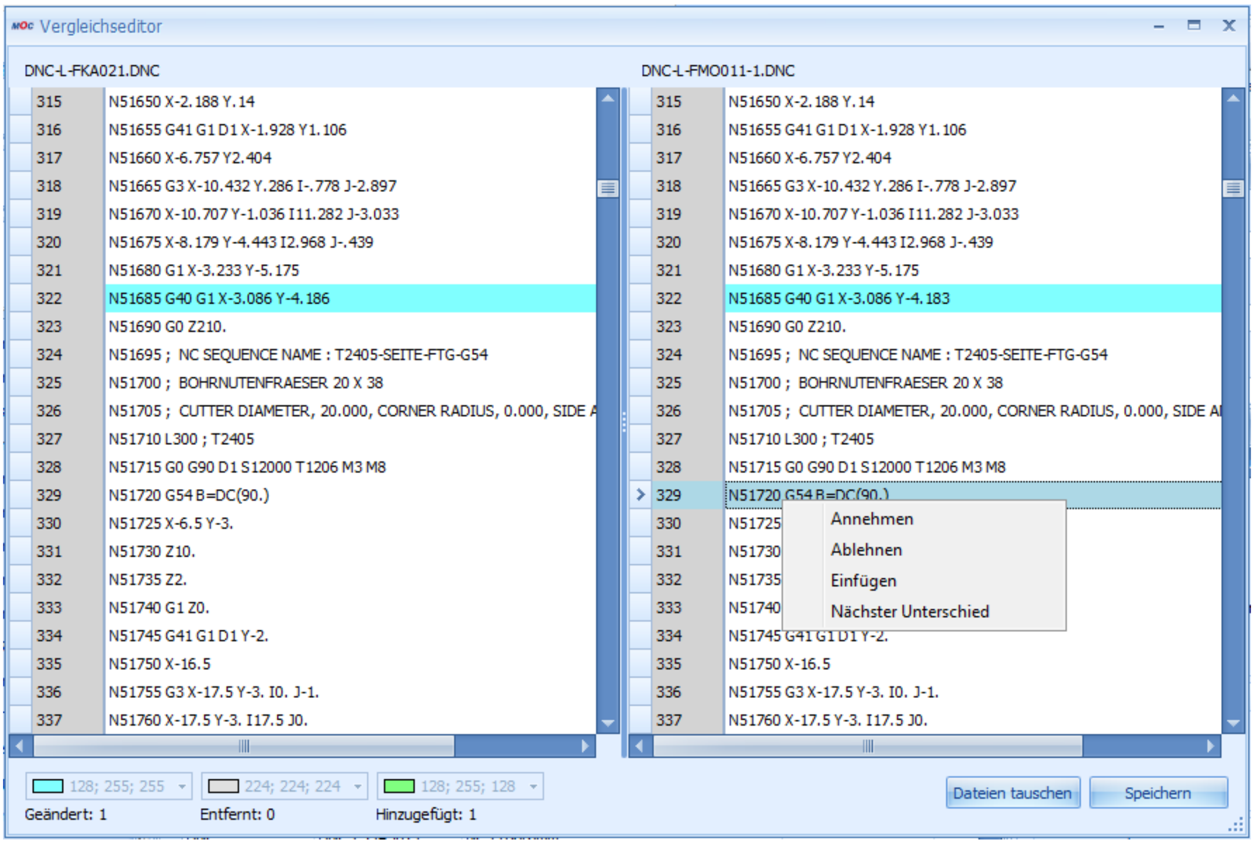
Opens the dialog to change a resource status.



Functions – Release of resource

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right clicking) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. If you only select one resource, this button is not available as in this case you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration changes

Restart the terminal which the workplace/machine is assigned to in order for the terminal program to interpret the configurations or modifications made to this workplace/machine.

Deleting a machine/ workplace

In a first step, the system shows a confirmation prompt asking if you really want to delete the machine. If you confirm this prompt, the system makes an attempt to delete the workplace. You can only delete a workplace successfully, if:

- you have not yet collected data for the workplace;
- you have currently not assigned the workplace to a terminal or a line;
- you have currently not logged on operations to the workplace;
- you have not planned operations for the workplace.

If you delete the workplace successfully, the system also deletes all configuration data, e.g. status assignments, for this workplace.

Checking Business Parameter Containers (BSCs)

See for further details on how to check the system against business parameters.